

# GapMind prediction (in silico)

Pathway predicted present (+)

Pathway predicted absent (—)

Experimental outcome  
(in vivo)

Growth observed (+)

No growth (—)

## a TRUE POSITIVE

concordant

### BIOLOGICAL STATE

Genes present and functional  
Pathway correctly annotated as present  
Experiment accurately measures growth

### INTERPRETATION

Prediction and phenotype agree:  
gene-to-function mapping is supported.

High-quality training sample

## b FALSE NEGATIVE

discordant

### POSSIBLE CAUSES

- **ANNOTATION** homolog not detected
- **BIOLOGY** alternative pathway present
- **BIOLOGY** promiscuous enzyme activity
- **MEASUREMENT** false-positive growth call
- **MEDIA** growth on base media (carryover)

Growth via unknown mechanism

## c FALSE POSITIVE

discordant

### POSSIBLE CAUSES

- **REGULATION** pathway repressed under condition
- **MEDIA** insufficient incubation time
- **MEDIA** missing cofactors / inducers
- **ANNOTATION** false-positive homology call
- **BIOLOGY** incomplete pathway (missing step)

Genes present but not functional

## d TRUE NEGATIVE

concordant

### BIOLOGICAL STATE

Genes absent from genome  
Pathway correctly annotated as absent  
No alternative pathway or leaky activity  
Experiment accurately measures no-growth

### INTERPRETATION

Absence of capability confirmed by both  
genome content and phenotype.

High-quality training sample